

SymPy Bug Card

Bug 13: nonlinsolve accepts an undefined point as a solution

Task. Solve the rational system with SymPy's `nonlinsolve`.

$$\left\{ \frac{xy - x}{x - 1} - y = 0, \quad y - 1 = 0 \right\}$$

SymPy's answer.

$$\text{nonlinsolve}\left(\left[\frac{xy - x}{x - 1} - y, y - 1\right], [x, y]\right) = \{(1, 1)\}.$$

Correct answer.

$$\emptyset$$

Explanation. Substituting $(1, 1)$ makes the original denominator $x - 1$ equal to 0, so the first rational equation is undefined there. Clearing the denominator yields the polynomial condition $y - x = 0$, but that step is valid only when $x \neq 1$; its apparent solution $(1, 1)$ is therefore removed by the original domain restriction.

Likely root cause. In `sympy/solvers/solveset.py`, `nonlinsolve` records denominators before converting rational equations to numerators, but its no-remaining-equations polynomial success path returns `FiniteSet(*map(to_tuple, poly_sol))` before applying those denominator exclusions. This is a new failure mode with no public precedent.

Artifact (GitHub).

[bug-report/bug-013-nonlinsolve-accepts-an-undefined-point-as-a-solution/](#)